

BLUE EIGHTY-EIGHT

FOH CHANNEL PROCESSING — DiGiCo Quantum 225 · Fountain Square · 2026-06-20

17 active FOH channels · 2 monitor-only · Outdoor · L-Acoustics A15/KS21 · Cincinnati, OH

INPUT LIST — ACTIVE CHANNELS

Channels with no mic/DI listed are excluded. Ch 23 and 24 are monitor-only — no FOH processing, no EQ cards.

CH	SOURCE	MIC / DI	48V
9	E-Kit L	DI	—
10	E-Kit R	DI	—
11	Bass DI	DI	—
12	Bass Keys	DI	—
13	Guitar 1	Sennheiser e609	—
14	Guitar 2	Sennheiser e609	—
17	Keys	DI	—
18	Sax 1	AT Pro 35	48V
19	Sax 2	AT Pro 35	48V
20	Perc 1 (Bongos)	Shure SM57	—
21	Perc 2 (Toys)	Shure SM57	—
25	Wayne (vox)	Shure SM58	—
26	Brent (vox)	Shure SM58	—
27	Chris (vox)	Shure SM58	—
28	John (vox)	Shure SM58	—
29	Carrie (vox)	Shure SM58	—
30	MD (vox)	Shure SM58	—
23	Click (<i>monitor-only</i>)	DI	—
24	Guide (<i>monitor-only</i>)	DI	—

CONSOLE REFERENCE

Channel Labels

CH	SHORT LABEL	INSTRUMENT	MIC / DI	48V	HPF Hz	VCA GROUP
9	EKIT L	E-Kit L	DI	—	40	E-KIT
10	EKIT R	E-Kit R	DI	—	40	E-KIT
11	BASS DI	Bass DI	DI	—	40	BASS
12	BASS KEYS	Bass Keys	DI	—	80	BASS
13	GTR 1	Guitar 1	Sennheiser e609	—	120	GUITARS
14	GTR 2	Guitar 2	Sennheiser e609	—	120	GUITARS
17	KEYS	Keys	DI	—	80	KEYS/MISC

18	SAX 1	Sax 1	AT Pro 35	48V	150	KEYS/MISC
19	SAX 2	Sax 2	AT Pro 35	48V	150	KEYS/MISC
20	PERC 1	Perc 1 (Bongos)	Shure SM57	—	150	KEYS/MISC
21	PERC 2	Perc 2 (Toys)	Shure SM57	—	200	KEYS/MISC
25	WAYNE	Wayne (vox)	Shure SM58	—	120	VOCALS
26	BRENT	Brent (vox)	Shure SM58	—	120	VOCALS
27	CHRIS	Chris (vox)	Shure SM58	—	120	VOCALS
28	JOHN	John (vox)	Shure SM58	—	120	VOCALS
29	CARRIE	Carrie (vox)	Shure SM58	—	120	VOCALS
30	MD	MD (vox)	Shure SM58	—	120	VOCALS
23	CLICK	Click	DI	—	—	Monitor-only
24	GUIDE	Guide	DI	—	—	Monitor-only

VCA Groups

VCA	NAME	CHANNELS
1	E-KIT	Ch 9, 10
2	BASS	Ch 11, 12
3	GUITARS	Ch 13, 14
4	KEYS/MISC	Ch 17, 18, 19, 20, 21
5	VOCALS	Ch 25, 26, 27, 28, 29, 30
6	ALL	VCA 1–5 master

WATCHPOINTS

⚠ PHANTOM POWER — Ch 18 & 19 (AT Pro 35)

Both sax channels require 48V phantom. The AT Pro 35 runs through an AT8538 inline phantom power module — no phantom means no signal, not degraded signal. Confirm phantom is on before soundcheck. All other channels are passive DIs or dynamic mics; no phantom required elsewhere on this show.

⚠ AT PRO 35 MIC HPF SWITCH — Ch 18 & 19

The AT Pro 35 has a built-in 80 Hz HPF switch at the capsule. Engage it on both mics before soundcheck. The console HPF sits above it at 150 Hz — the combination is intentional and handles outdoor wind and stage handling noise. Gate both channels; key slap between phrases is constant on clip-on sax mics.

E-KIT STEREO PAIR — Ch 9 & 10

E-kick is folded into the stereo L/R output from the drum module. HPF stays at 40 Hz on both channels — anything higher eats kick body. Link Ch 9 and Ch 10 as a stereo pair on the Q225. Match gain before any EQ. Polarity check both channels against each other in mono. Kit image should sit naturally in the center field before soundcheck opens.

MD DOUBLING — Ch 12, 17, 30

The Music Director is on three channels: Ch 12 (Bass Keys), Ch 17 (Keys), and Ch 30 (vocal). This is the most routing-complex position on the stage. Coordinate with the M32 operator before soundcheck — the MD monitor mix needs to account for all three sources. The vocal channel (Ch 30) is the primary FOH presence.

FSQ OUTDOOR EQ APPROACH

All channels on this show follow the FSQ outdoor EQ rule: cuts over boosts throughout. The L-Acoustics A15/KS21 rig has clean high-end and deep sub extension — it doesn't need help from channel EQ. Vocals are cuts-only with no exceptions. The primary risk is gain-before-feedback on open mics in a downtown ambient noise floor, not room reflections.

TWO GUITARS — Ch 13 & 14 CARVE

Both guitar channels use the same mic (e609). Starting values are intentionally offset — Ch 13 cuts at 3 kHz / 300 Hz, Ch 14 cuts at 3.5 kHz / 350 Hz — to create natural frequency separation without forcing it. If they compete in the mix, widen the separation before reaching for level. Check on the guitar bus together rather than independently.

MONITOR-ONLY — Ch 23 & 24

Click (Ch 23) and Guide (Ch 24) are routed to monitors only. Confirm no FOH path is open on either channel. These feeds originate from playback — confirm routing at patch and verify at the M32 before soundcheck.

BLUE EIGHTY-EIGHT

FOH CHANNEL PROCESSING — DiGiCo Q225 · Fountain Square · 2026-06-20

Band order: **Band 4** (highest) → **Band 3** → **Band 2** → **Band 1** (lowest) — matches Q225 console display.
HPFs marked * are higher than indoor equivalents; outdoor ambient and wind justify the offset.

Ch 9 — E-Kit L · DI		
Section	Parameter / Value	Details
HPF	40 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 8 kHz	Q 0.71; Shelf — Tame digital top-end glare from the drum module. The module already processes the signal; this smooths the HF character.
Band 3; Upper-Mid Bell	-3 dB @ 3 kHz	Q 2.0 — Module mid-forward response. A cut here cleans up the slightly synthetic midrange without touching the fundamental.
Band 2; Lower-Mid Bell	-3 dB @ 250 Hz	Q 2.0 — Low-mid mud. Present in most DI sources; static cut is sufficient outdoors.
Band 1; Low	FLAT	HPF at 40 Hz handles the bottom. E-kick is in this stereo mix — don't choke the fundamental.
Quick Summary		E-kit L/R stereo output from module — signal arrives pre-processed. HPF stays low (40 Hz) because the e-kick is folded into this stereo mix; anything higher starts eating kick body. Match gain Ch 9 and Ch 10 before any EQ. Polarity check both against each other in mono before soundcheck.

Ch 10 — E-Kit R · DI		
Section	Parameter / Value	Details
HPF	40 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 8 kHz	Q 0.71; Shelf — Match Ch 9 exactly.
Band 3; Upper-Mid Bell	-3 dB @ 3 kHz	Q 2.0 — Match Ch 9.
Band 2; Lower-Mid Bell	-3 dB @ 250 Hz	Q 2.0 — Match Ch 9.
Band 1; Low	FLAT	Match Ch 9.
Quick Summary		Mirror Ch 9 exactly. Link as a stereo pair on the Q225 before soundcheck and adjust as a unit. Confirm stereo image sits naturally in the field — nudge pan if the kit reads asymmetric.

Ch 11 — Bass DI · DI

Section	Parameter / Value	Details
HPF	40 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 5 kHz	Q 0.71; Shelf — DI glare and string noise above the useful bass range. The A15/KS21 rig is clean up top; a shelf cut tidies the source without touching character.
Band 3; Upper-Mid Bell	-2 dB @ 800 Hz	Q 2.0 — Mid-bass honk. Common in active bass DIs; a modest cut opens up the mix without hollowing the fundamental.
Band 2; Lower-Mid Bell	-3 dB @ 250 Hz	Q 2.0 — Mud control. This range accumulates across the mix outdoors; address it on every DI source.
Band 1; Low	FLAT	The A15/KS21 sub arch has deep extension — don't pile on. Let the fundamental sit naturally.
Quick Summary		Electric bass DI. FSQ PA (A15/KS21) has serious low-end extension and doesn't need help — resist adding low shelf here. Cut the mud at 250 Hz, knock back the honk at 800 Hz, and let the fundamental breathe. Carve space between this and Bass Keys (Ch 12) once both are open.

Ch 12 — Bass Keys · DI

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 6 kHz	Q 0.71; Shelf — Keyboard DI is usually clean; this trims the digital top end.
Band 3; Upper-Mid Bell	-2 dB @ 1 kHz	Q 2.0 — Keyboard mid-push at 1 kHz is the primary culprit when keys compete with bass DI. A modest cut creates space.
Band 2; Lower-Mid Bell	-3 dB @ 250 Hz	Q 2.0 — Same mud frequency as Ch 11. Treat it the same way.
Band 1; Low	FLAT	HPF at 80 Hz. Let it sit — this is keyboard bass, not sub bass.
Quick Summary		Keyboard bass DI. Primary concern is carving against Ch 11 (electric bass) — they live in the same frequency range. The 1 kHz cut creates the most useful separation. This is also the MD, who is simultaneously on Ch 17 (Keys) and Ch 30 (vocal) — plan the monitor mix before soundcheck.

Ch 13 — Guitar 1 · Sennheiser e609

Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 8 kHz	Q 0.71; Shelf — e609 on a guitar cab has a forward upper-mid character. A gentle shelf above 8 kHz tames harshness without dulling the tone.
Band 3; Upper-Mid Bell	-3 dB @ 3 kHz	Q 2.0 — The e609 natural presence peak. Cut rather than boost — let the mic's character work without adding more edge.
Band 2; Lower-Mid Bell	-4 dB @ 300 Hz	Q 2.0 — Cabinet body buildup. Most common problem area on e609-on-cab; needs addressing on every guitar channel outdoors.
Band 1; Low Bell	-4 dB @ 150 Hz	Q 1.5 — Low-mid rumble below the guitar's useful fundamental. Outdoors this range accumulates from stage floor coupling.
Quick Summary		Guitar 1, e609 side-address over the cab face. The 609 is honest — what the amp sounds like is what you get. Keep cuts clean: 300 Hz for cab body, 150 Hz to clean up the floor. The 3 kHz cut controls the 609 presence peak rather than fighting it. Two e609 channels (13 and 14) — if they compete in the mix, widen the cut frequencies before reaching for level.

Ch 14 — Guitar 2 · Sennheiser e609

Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 8 kHz	Q 0.71; Shelf — Match Ch 13 start, adjust per amp tone.
Band 3; Upper-Mid Bell	-3 dB @ 3.5 kHz	Q 2.0 — Shifted slightly higher than Ch 13 to create natural carving between the two guitar channels.
Band 2; Lower-Mid Bell	-4 dB @ 350 Hz	Q 2.0 — Cabinet body. Shifted slightly from Ch 13 (300 Hz) to differentiate the two guitars.
Band 1; Low Bell	-4 dB @ 150 Hz	Q 1.5 — Match Ch 13.
Quick Summary		Guitar 2, e609. Same mic and approach as Ch 13 — small offsets in Band 3 (3.5 kHz vs 3 kHz) and Band 2 (350 Hz vs 300 Hz) create natural frequency separation between the two guitars. Adjust once both are open. If one guitar is significantly brighter or darker than the other, go back to matching and adjust by ear rather than forcing separation.

Ch 17 — Keys · DI

Section	Parameter / Value	Details
HPF	80 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Light trim at the top. Keyboard DI doesn't need extended air in a live outdoor mix.
Band 3; Upper-Mid Bell	-2 dB @ 3 kHz	Q 2.0 — Keyboard presence peak. A modest cut creates room for vocals above without dulling the instrument.
Band 2; Lower-Mid Bell	-3 dB @ 250 Hz	Q 2.0 — Standard mud control on all DI sources.
Band 1; Low	FLAT	HPF at 80 Hz. Let the left hand sit naturally; the subwoofer arch handles the bottom.
Quick Summary		Keys DI — role TBD (Rhodes/synth/piano). MD is simultaneously on Ch 12 (Bass Keys) and Ch 30 (vocal) — plan monitor complexity before soundcheck. If the keys are covering piano parts, the 3 kHz cut may need to come back slightly to keep attack. If covering pads or strings, keep as-is.

Ch 18 — Sax 1 · AT Pro 35 ⚠ 48V

Section	Parameter / Value	Details
HPF	150 Hz @ 24 dB/Oct	* HPF higher than indoor equivalent — outdoor wind noise and ambient floor justify the offset.
Band 4; High Shelf	-3 dB @ 10 kHz	Q 0.71; Shelf — The Pro 35 response falls off above 15 kHz anyway. Rolling off at 10 kHz removes harshness in the cap rolloff region without losing useful content. Do not try to boost air on this mic — it isn't there.
Band 3; Upper-Mid Bell	-3 dB @ 1.2 kHz	Q 2.5; narrow — Clip-on key noise. Mechanical transients from pad mechanisms sit in the 1–1.5 kHz range and bleed constantly between phrases. Narrow cut minimises impact on tone.
Band 2; Lower-Mid Bell	-3 dB @ 350 Hz	Q 2.0 — Boxy proximity buildup from the clip-on position near the bell. Close placement exaggerates this range.
Band 1; Low	FLAT	Console HPF at 150 Hz handles this plus the mic built-in 80 Hz HPF. Don't stack more low cuts.
Quick Summary		AT Pro 35 clip-on on sax. Two mandatory checks before soundcheck: (1) 48V phantom is on — the Pro 35 runs off the AT8538 inline module and requires phantom; no phantom means no signal; (2) the mic built-in 80 Hz HPF switch is engaged at the capsule. The 1.2 kHz narrow cut is the most important move on this mic — key slap between phrases is the constant enemy on clip-on sax. Gate strongly recommended; set threshold so it closes cleanly between phrases without chopping held notes.

Ch 19 — Sax 2 · AT Pro 35 △ 48V

Section	Parameter / Value	Details
HPF	150 Hz @ 24 dB/Oct	* HPF higher than indoor equivalent — outdoor wind noise and ambient floor justify the offset.
Band 4; High Shelf	-3 dB @ 10 kHz	Q 0.71; Shelf — Match Ch 18 as starting point.
Band 3; Upper-Mid Bell	-3 dB @ 1.2 kHz	Q 2.5 — Match Ch 18. Adjust independently once both sax players are open.
Band 2; Lower-Mid Bell	-3 dB @ 350 Hz	Q 2.0 — Match Ch 18.
Band 1; Low	FLAT	Match Ch 18.
Quick Summary		Mirror Ch 18 as the starting point, then adjust independently per player. Different clip positions and instrument voices will need individual tuning — same mic, same approach, different ears. Same phantom and mic HPF requirements apply. Gate independently of Ch 18.

Ch 20 — Perc 1 (Bongos) · Shure SM57

Section	Parameter / Value	Details
HPF	150 Hz @ 24 dB/Oct	* HPF higher than indoor equivalent — outdoor wind noise and ambient floor justify the offset.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Keep the SM57 from getting harsh on drum transients.
Band 3; Upper-Mid Bell	-3 dB @ 3 kHz	Q 2.0 — SM57 presence peak on a percussion source. Cut rather than fight it.
Band 2; Lower-Mid Bell	-3 dB @ 350 Hz	Q 2.0 — Hand drum body buildup. Bongo fundamentals sit above this; the cut cleans up without thinning the tone.
Band 1; Low Bell	-4 dB @ 200 Hz	Q 1.5 — Clear out low-mid below bongo fundamental. Bongos don't produce useful content here; the cut opens space for bass and kick.
Quick Summary		SM57 on bongos. Tight cardioid keeps bleed manageable on an outdoor stage with multiple open mics. HPF at 150 Hz is higher than you'd use indoors; outdoor ambient noise and floor coupling justify it. The 3 kHz cut manages the SM57 presence peak — don't boost over it. Bongo attack sits in the 3–5 kHz range naturally; let the mic provide it.

Ch 21 — Perc 2 (Toys) · Shure SM57		
Section	Parameter / Value	Details
HPF	200 Hz @ 24 dB/Oct	* HPF higher than indoor equivalent — outdoor wind noise and ambient floor justify the offset.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — SM57 top-end management.
Band 3; Upper-Mid Bell	-3 dB @ 3 kHz	Q 2.0 — SM57 presence peak. Same cut as Ch 20.
Band 2; Lower-Mid Bell	-4 dB @ 400 Hz	Q 2.0 — Percussion toys produce little useful content below 400 Hz; clear out the low-mid floor.
Band 1; Low	FLAT	HPF at 200 Hz handles this. Toys don't need low-end.
Quick Summary		SM57 on toys — percussion of unknown type. HPF at 200 Hz because most small percussion instruments produce nothing useful below that point. All cuts, no boosts. Adjust at soundcheck once the actual instruments are visible. If shakers are in play, the 3–4 kHz cut may need to come back slightly to let shimmer through.

Ch 25 — Wayne (vox) · Shure SM58		
Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Outdoor vocal air management. SM58 handles this range naturally; light shelf trim stays out of the feedback zone without dulling intelligibility.
Band 3; Upper-Mid Bell	-3 dB @ 2.5 kHz	Q 1.5 — SM58 upper-mid harshness and primary feedback frequency outdoors. Cuts only per FSQ vocal rule.
Band 2; Lower-Mid Bell	-4 dB @ 250 Hz	Q 2.0 — SM58 proximity effect management. More aggressive than indoors; outdoor proximity buildup and wind can exaggerate this range significantly.
Band 1; Low	FLAT	HPF at 120 Hz. SM58 proximity effect is real; the HPF and the 250 Hz cut together handle the low-mid buildup.
Quick Summary		SM58, lead vocal. FSQ vocal rule: cuts only, no boosts. The SM58 outdoors means more proximity effect than you'd see at Memo; the 250 Hz cut at -4 dB is heavier than an indoor starting point for that reason. The 2.5 kHz cut manages harshness and keeps feedback margin comfortable. Don't chase brightness by boosting — solve it at mic placement if needed. Adjust per Wayne's voice at soundcheck.

Ch 26 — Brent (vox) · Shure SM58

Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Match Ch 25 start.
Band 3; Upper-Mid Bell	-3 dB @ 2.5 kHz	Q 1.5 — Match Ch 25. Adjust per voice.
Band 2; Lower-Mid Bell	-4 dB @ 250 Hz	Q 2.0 — Match Ch 25.
Band 1; Low	FLAT	Match Ch 25.
Quick Summary		SM58, vocal — Brent. Start at Ch 25 values, adjust per voice at soundcheck. Same outdoor proximity and feedback management logic applies.

Ch 27 — Chris (vox) · Shure SM58

Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Match Ch 25.
Band 3; Upper-Mid Bell	-3 dB @ 2.5 kHz	Q 1.5 — Match Ch 25.
Band 2; Lower-Mid Bell	-4 dB @ 250 Hz	Q 2.0 — Match Ch 25.
Band 1; Low	FLAT	Match Ch 25.
Quick Summary		SM58, vocal — Chris. Start at Ch 25 values and adjust per voice.

Ch 28 — John (vox) · Shure SM58

Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Match Ch 25.
Band 3; Upper-Mid Bell	-3 dB @ 2.5 kHz	Q 1.5 — Match Ch 25.
Band 2; Lower-Mid Bell	-4 dB @ 250 Hz	Q 2.0 — Match Ch 25.
Band 1; Low	FLAT	Match Ch 25.
Quick Summary		SM58, vocal — John. Start at Ch 25 values and adjust per voice.

Ch 29 — Carrie (vox) · Shure SM58		
Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Match Ch 25.
Band 3; Upper-Mid Bell	-3 dB @ 2.5 kHz	Q 1.5 — Match Ch 25. If sibilance is an issue, add a narrow -2 to -3 dB cut at 6–7 kHz — stays within cuts-only rule.
Band 2; Lower-Mid Bell	-4 dB @ 250 Hz	Q 2.0 — Match Ch 25.
Band 1; Low	FLAT	Match Ch 25.
Quick Summary		SM58, vocal — Carrie. Start at Ch 25. If sibilance is an issue, add a narrow -2 to -3 dB cut at 6–7 kHz; no conflict with the cuts-only vocal rule. Adjust per voice at soundcheck.

Ch 30 — MD (vox) · Shure SM58		
Section	Parameter / Value	Details
HPF	120 Hz @ 24 dB/Oct	Remove sub-bass stage noise.
Band 4; High Shelf	-2 dB @ 10 kHz	Q 0.71; Shelf — Match Ch 25.
Band 3; Upper-Mid Bell	-3 dB @ 2.5 kHz	Q 1.5 — Match Ch 25.
Band 2; Lower-Mid Bell	-4 dB @ 250 Hz	Q 2.0 — Match Ch 25.
Band 1; Low	FLAT	Match Ch 25.
Quick Summary		SM58, vocal — MD (Music Director). MD is simultaneously on Ch 17 (Keys) and Ch 12 (Bass Keys) — this is the most routing-complex position. Plan monitor complexity before soundcheck and coordinate with the M32 operator. Confirm that only the vocal channel (Ch 30) has a FOH path on the MD's channels unless the Keys and Bass Keys are mixed differently.

FILE INDEX	
Folder: Fountain Square / 2026-06-20 Blue Eighty-Eight /	
Blue Eighty-Eight - FOH Channel Processing.md	Machine-readable EQ source — patcher input data, all active channels
Blue Eighty-Eight - FOH Channel Processing.html	This document (source)
Blue Eighty-Eight - FOH Channel Processing.pdf	PDF render — master FOH reference for load-in
apply_blue88.py	Q225 .ses patcher script — pending Stage 6
Blue Eighty-Eight.ses	DiGiCo Q225 show file — pending Stage 6